

Лабораторная работа № 10

Расширенная настройка SMTP-сервера

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Цель:

Настроить интеграцию Postfix с Dovecot, LMTP, SMTP AUTH и TLS.



Сокет Dovecot LMTP

```
root@server:~  
root@server:~# nano /etc/dovecot/dovecot.conf  
root@server:~# nano /etc/dovecot/conf.d/10-master.conf  
root@server:~# grep -A6 '^service lmtp' /etc/dovecot/conf.d/10-master.conf  
service lmtp {  
    unix_listener /var/spool/postfix/private/dovecot-lmtp {  
        group = postfix  
        user = postfix  
        mode = 0600  
    }  
}
```

Смысл результата

В Dovecot включён протокол LMTP и создан сокет внутри каталога Postfix. Через него MTA передаёт письма локальному агенту доставки.

Передача доставки в LMTP

```
root@server:~# systemctl restart postfix
systemctl restart dovecot
root@server:~# systemctl status postfix --no-pager
systemctl status dovecot --no-pager
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-09-06 10:38:35 UTC; 15s ago
   Invocation: bab4480002e84e70ac8e79d01f1acb55
   Process: 17934 ExecStartPre=/usr/sbin/restorecon -R /var/spool/postfix/pid (code=exited, status=0/SUCCESS)
   Process: 17935 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=exited, status=0/SUCCESS)
   Process: 17938 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, status=0/SUCCESS)
   Process: 17941 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCESS)
   Main PID: 18010 (master)
   Tasks: 3 (limit: 10232)
   Memory: 2.1M (peak: 3.4M)
   CPU: 537ms
   CGroup: /system.slice/postfix.service
           └─18010 /usr/libexec/postfix/master -w
             └─18012 pickup -l -t unix -u
               └─18013 qmgr -l -t unix -u

Sep 06 10:38:34 server systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Sep 06 10:38:35 server postfix[18006]: postfix/postlog: starting the Postfix mail system
Sep 06 10:38:35 server postfix/postfix-script[18006]: starting the Postfix mail system
Sep 06 10:38:35 server postfix/master[18010]: daemon started -- version 3.8.5, configuration /etc/postfix
Sep 06 10:38:35 server systemd[1]: Started postfix.service - Postfix Mail Transport Agent.
Hint: Some lines were ellipsized, use -l to show in full.
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-09-06 10:38:35 UTC; 15s ago
   Invocation: ef1deb265c524cbf82814ac8772b7094
   Docs: man:dovecot(1)
         https://doc.dovecot.org/
   Main PID: 18019 (dovecot)
   Status: "v2.3.21 (47349e2482) running"
   Tasks: 4 (limit: 10232)
   Memory: 5.8M (peak: 6.1M)
   CPU: 91ms
   CGroup: /system.slice/dovecot.service
           └─18019 /usr/sbin/dovecot -F
             └─18020 dovecot/anvil
               └─18021 dovecot/log
                 └─18022 dovecot/config

Sep 06 10:38:35 server systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email server...
Sep 06 10:38:35 server dovecot[18019]: master: Dovecot v2.3.21 (47349e2482) starting up for imap,..., lmtp
Sep 06 10:38:35 server systemd[1]: Started dovecot.service - Dovecot IMAP/POP3 email server.
Hint: Some lines were ellipsized, use -l to show in full.
root@server:~#
```

Смысл результата

В main.cf Postfix параметр mailbox_transport переключён на dovecot-lmtp. Конфигурация связывает приём SMTP с хранением Dovecot.

Успешная LMTP-доставка

```
root@server:~  
ent.ndbaranov.net>  
Sep  6 10:39:43 server postfix/smtpd[18039]: disconnect from unknown[192.168.1.31] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7  
Sep  6 10:39:43 server postfix/qmgr[18013]: 6FF4560D7316: from=<ndbaranov@client.ndbaranov.net>, size=541, nrcpt=1 (queue active)  
Sep  6 10:39:43 server postfix/local[18044]: 6FF4560D7316: passing <ndbaranov@ndbaranov.net> to transport=lmtp  
Sep  6 10:39:43 server dovecot[18021]: lmtp(18046): Connect from local  
Sep  6 10:39:43 server postfix/lmtp[18045]: 6FF4560D7316: to=<ndbaranov@ndbaranov.net>, relay=server.ndbaranov.net[private:dovecot-lmtp], delay=0.18, delays=0.04/0.02/0.06/0.06, dsn=5.1.1, status=bounced (host server.ndbaranov.net[private:dovecot-lmtp] said: 550 5.1.1 <ndbaranov@ndbaranov.net> User doesn't exist: ndbaranov@ndbaranov.net (in reply to RCPT TO command))  
Sep  6 10:39:43 server dovecot[18021]: lmtp(18046): Disconnect from local: Logged out (state=READY)  
Sep  6 10:39:43 server postfix/cleanup[18043]: 979B560D7E99: message-id=<20260906103943.979B560D7E99@server.ndbaranov.net>  
Sep  6 10:39:43 server postfix/qmgr[18013]: 979B560D7E99: from=<>, size=2750, nrcpt=1 (queue active)  
Sep  6 10:39:43 server postfix/bounce[18050]: 6FF4560D7316: sender non-delivery notification: 979B560D7E99  
Sep  6 10:39:43 server postfix/qmgr[18013]: 6FF4560D7316: removed  
Sep  6 10:39:46 server postfix/smtp[18051]: connect to client.ndbaranov.net[192.168.1.30]:25: No route to host  
Sep  6 10:39:46 server postfix/smtp[18051]: 979B560D7E99: to=<ndbaranov@client.ndbaranov.net>, relay=none, delay=3.1, delays=0.01/0.04/3.1/0, dsn=4.4.1, status=deferred (connect to client.ndbaranov.net[192.168.1.30]:25: No route to host)  
Sep  6 10:42:19 server postfix/smtpd[18099]: connect from unknown[192.168.1.31]  
Sep  6 10:42:19 server postfix/smtpd[18099]: DF1D560D7316: client=unknown[192.168.1.31]  
Sep  6 10:42:19 server postfix/cleanup[18102]: DF1D560D7316: message-id=<20260906104219.B83A320E160A@client.ndbaranov.net>  
Sep  6 10:42:19 server postfix/smtpd[18099]: disconnect from unknown[192.168.1.31] ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7  
Sep  6 10:42:19 server postfix/qmgr[18013]: DF1D560D7316: from=<ndbaranov@client.ndbaranov.net>, size=541, nrcpt=1 (queue active)  
Sep  6 10:42:19 server postfix/local[18103]: DF1D560D7316: passing <ndbaranov@ndbaranov.net> to transport=lmtp  
Sep  6 10:42:19 server dovecot[18021]: lmtp(18105): Connect from local  
Sep  6 10:42:20 server postfix/lmtp[18104]: DF1D560D7316: to=<ndbaranov@ndbaranov.net>, relay=server.ndbaranov.net[private:dovecot-lmtp], delay=0.13, delays=0.04/0.02/0.04/0.04, dsn=5.1.1, status=bounced (host server.ndbaranov.net[private:dovecot-lmtp] said: 550 5.1.1 <ndbaranov@ndbaranov.net> User doesn't exist: ndbaranov@ndbaranov.net (in reply to RCPT TO command))  
Sep  6 10:42:20 server dovecot[18021]: lmtp(18105): Disconnect from local: Logged out (state=READY)  
Sep  6 10:42:20 server postfix/cleanup[18102]: 0615F60D7EA8: message-id=<20260906104220.0615F60D7EA8@server.ndbaranov.net>  
Sep  6 10:42:20 server postfix/bounce[18107]: DF1D560D7316: sender non-delivery notification: 0615F60D7EA8  
Sep  6 10:42:20 server postfix/qmgr[18013]: 0615F60D7EA8: from=<>, size=2750, nrcpt=1 (queue active)  
Sep  6 10:42:20 server postfix/qmgr[18013]: DF1D560D7316: removed  
Sep  6 10:42:23 server postfix/smtp[18108]: connect to client.ndbaranov.net[192.168.1.30]:25: No route to host  
Sep  6 10:42:23 server postfix/smtp[18108]: 0615F60D7EA8: to=<ndbaranov@client.ndbaranov.net>, relay=none, delay=3.1, delays=0.01/0.03/3.1/0, dsn=4.4.1, status=deferred (connect to client.ndbaranov.net[192.168.1.30]:25: No route to host)
```

Смысл результата

В почтовом журнале для тестового сообщения указан status=sent и транспорт LMTP. Это подтверждает прохождение письма через новый маршрут.

```
ndbaranov.net[private/dovecot-lmtp], delay=0.15, delays=0.03/0.02/0.04/0.06, dsn=2.0.0, status=sent (250 2.0.0 <ndbaranov@ndbaranov.net> Ly3xIAREnWr8RgAAot0D+g Saved)
Sep  6 10:44:20 server postfix/qmgr[18013]: 74B1060D7EAF: removed
Sep  6 10:44:20 server dovecot[18158]: lmtp(18172): Disconnect from local: Logged out (state=READY)
```

Смысл результата

Файл доставленного сообщения появился в каталоге Maildir пользователя. Содержимое заголовков соответствует отправленному тесту.

Auth-сокеты для Postfix

```
ndbaranov@server:~  
root@server:~# su - ndbaranov  
Last login: Sun Sep  6 10:04:47 UTC 2026 on pts/1  
[ndbaranov@server ~]$ MAIL=~/.Maildir/ mail  
s-nail version v14.9.24.  Type `?' for help  
/home/ndbaranov/Maildir: 3 messages 1 new 2 unread  
  1 Super User      2026-09-06 10:02  17/664  "IMAP test 1  
  U 2 Super User      2026-09-06 10:02  17/666  "IMAP test 3  
  N 3 ndbaranov@client.ndb 2026-09-06 10:44  21/811  "LMTP test  
& q  
Held 3 messages in /home/ndbaranov/Maildir  
[ndbaranov@server ~]$
```

Смысл результата

Dovecot создаёт private/auth с правами пользователя postfix. Этот сокет используется SMTP-сервером для проверки имени и пароля.

Включение SMTP AUTH

```
root@server:~  
[ndbaranov@server ~]$ suo -i  
bash: suo: command not found...  
[ndbaranov@server ~]$ sudo -i  
[sudo] password for ndbaranov:  
Sorry, try again.  
[sudo] password for ndbaranov:  
[root@server ~]# nano /etc/dovecot/conf.d/10-master.conf  
[root@server ~]# doveconf -n >/dev/null  
[root@server ~]# grep -A15 '^service auth' /etc/dovecot/conf.d/10-master.conf  
service auth {  
  # auth_socket_path points to this userdb socket by default. It's typically  
  # used by dovecot-lda, doveadm, possibly imap process, etc. Users that have  
  # full permissions to this socket are able to get a list of all usernames and  
  # get the results of everyone's userdb lookups.  
  #  
  # The default 0666 mode allows anyone to connect to the socket, but the  
  # userdb lookups will succeed only if the userdb returns an "uid" field that  
  # matches the caller process's UID. Also if caller's uid or gid matches the  
  # socket's uid or gid the lookup succeeds. Anything else causes a failure.  
  #  
  # To give the caller full permissions to lookup all users, set the mode to  
  # something else than 0666 and Dovecot lets the kernel enforce the  
  # permissions (e.g. 0777 allows everyone full permissions).  
  unix_listener auth-userdb {  
    mode = 0600  
  }  
--  
service auth-worker {  
  # Auth worker process is run as root by default, so that it can access  
  # /etc/shadow. If this isn't necessary, the user should be changed to  
  # $default_internal_user.  
  #user = root  
}  
  
service dict {  
  # If dict proxy is used, mail processes should have access to its socket.  
  # For example: mode=0660, group=vmail and global mail_access_groups=vmail  
  unix_listener dict {  
    #mode = 0600  
    #user =  
    #group =  
  }  
}
```

Смысл результата

В Postfix активирован SASL через Dovecot и запрещена анонимная аутентификация. postfix показывает выбранный тип и путь сокета.

TLS-сертификаты

```
ndbaranov@client:~  
PS D:\work\ndbaranov\vagrant> vagrant ssh client  
==> client: The machine you're attempting to SSH into is configured to use  
==> client: password-based authentication. Vagrant can't script entering the  
==> client: password for you. If you're prompted for a password, please enter  
==> client: the same password you have configured in the Vagrantfile.  
vagrant@127.0.0.1's password:  
Last login: Sun Sep  6 09:54:24 2026 from 10.0.2.2  
vagrant@client:~$ su -ndbaranov  
su: invalid option -- 'n'  
Try 'su --help' for more information.  
vagrant@client:~$ su - ndbaranov  
Password:  
ndbaranov@client:~$ echo . | mail -s "LMTP test" ndbaranov@ndbaranov.net  
ndbaranov@client:~$ echo . | mail -s "LMTP test" ndbaranov@ndbaranov.net  
ndbaranov@client:~$ echo . | mail -s "LMTP test" ndbaranov@ndbaranov.net  
ndbaranov@client:~$ printf 'ndbaranov\x00ndbaranov\x00123456' | base64  
bmRiYXJhbm92AG5kYmFyYW5vdGxMjM0NTY=  
ndbaranov@client:~$ telnet server.ndbaranov.net 25  
Trying 192.168.1.1...  
Connected to server.ndbaranov.net.  
Escape character is '^]'.  
220 server.ndbaranov.net ESMTF Postfix  
AUTH PLAIN bmRiYXJhbm92AG5kYmFyYW5vdGxMjM0NTY=  
235 2.7.0 Authentication successful  
quit  
221 2.0.0 Bye  
Connection closed by foreign host.  
ndbaranov@client:~$ telnet server.ndbaranov.net 25  
Trying 192.168.1.1...  
Connected to server.ndbaranov.net.  
Escape character is '^]'.  
220 server.ndbaranov.net ESMTF Postfix  
EHLO test  
250-server.ndbaranov.net  
250-PIPELINING  
250-SIZE 10240000  
250-VRFY  
250-ETRN  
250-STARTTLS  
250-AUTH PLAIN  
250-ENHANCEDSTATUSCODES  
250-8BITMIME  
250-DSN  
250-SMTPUTF8  
250 CHUNKING
```

Смысл результата

Для SMTP указаны сертификат и закрытый ключ Dovecot, включён STARTTLS. Проверка конфигурации не выявила ошибок доступа к файлам.

Сервис submission

```
root@server:~  
[root@server ~]# cp /etc/pki/dovecot/certs/dovecot.pem /etc/pki/tls/certs/  
cp /etc/pki/dovecot/private/dovecot.pem /etc/pki/tls/private/  
[root@server ~]# postconf -e 'smtpd_tls_cert_file=/etc/pki/tls/certs/dovecot.pem'  
postconf -e 'smtpd_tls_key_file=/etc/pki/tls/private/dovecot.pem'  
postconf -e 'smtpd_tls_session_cache_database = btree:/var/lib/postfix/smtpd_scache'  
postconf -e 'smtpd_tls_security_level = may'  
postconf -e 'smtpd_tls_security_level = may'  
[root@server ~]# nano /etc/postfix/master.cf  
[root@server ~]# nano /etc/postfix/master.cf  
[root@server ~]# postfix check  
[root@server ~]# firewall-cmd --get-services  
firewall-cmd --add-service=smtp-submission  
firewall-cmd --add-service=smtp-submission --permanent  
firewall-cmd --reload  
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602  
anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareo  
s-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-export  
er ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector crat  
edb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls d  
ocker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger forema  
n foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli  
a-client ganglia-master git gitea gpsd grafana gre high-availability http http3 https ident imap imaps i  
perf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kib  
ana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub  
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler  
-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-net  
work llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp mon  
godb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbi  
os-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storag  
econsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus p  
rometheus-node-exporter proxy-dhcp proxy-http ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radi  
us radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client  
samba-dc sane settlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls  
snmptls-trap snmptrap socks spideroak-lansync spotify-sync squid ssdp ssh ssh-custom statsrv steam-lan-t  
ransfer steam-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn synct  
hing syncthing-gui syncthing-relay synergy syscomlan syslog syslog-ng syslog-tls telnet tentacle terrari  
a tftp tile38 tinc tor-socks transmission-client turn turns upnp-client vdsml vnc-server vrrp warpinator  
wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-di  
scovery-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent  
zabbix-java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier  
success  
success  
success  
[root@server ~]# systemctl restart postfix  
[root@server ~]# ss -ltnp | grep ':587'  
[root@server ~]#
```

Смысл результата

В master.cf включён submission на порту 587 с обязательной аутентификацией. Firewalld разрешает подключение почтовых клиентов к этому порту.

TLS-рукопожатие openssl

```
root@server:~  
[root@server ~]# doveconf -n >/dev/null  
systemctl restart dovecot  
systemctl restart postfix  
[root@server ~]# postfix check  
[root@server ~]# firewall-cmd --get-services  
firewall-cmd --add-service=smtp-submission  
firewall-cmd --add-service=smtp-submission --permanent  
firewall-cmd --reload  
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602  
anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareo  
s-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-export  
er ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector crat  
edb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls d  
ocker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger forema  
n foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli  
a-client ganglia-master git gitea gpsd grafana gre high-availability http http3 https ident imap imaps i  
perf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kib  
ana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub  
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler  
-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-net  
work llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp mon  
godb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbi  
os-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storag  
econsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus p  
rometheus-node-exporter proxy-dhcp proxy-http ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radi  
us radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client  
samba-dc sane settlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls  
snmptls-trap snmptrap socks spideroak-lansync spotify-sync squid ssdp ssh ssh-custom statsrv steam-lan-t  
ransfer steam-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn synct  
hing syncthing-gui syncthing-relay synergy syscomlan syslog syslog-ng syslog-tls telnet tentacle terrari  
a tftp tile38 tinc tor-socks transmission-client turn turns upnp-client vdsu vnc-server vrrp warpinator  
wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-di  
scovery-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-client xmpp-local xmpp-server zabbix-agent  
zabbix-java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier  
Warning: ALREADY_ENABLED: 'smtp-submission' already in 'public'  
success  
Warning: ALREADY_ENABLED: smtp-submission  
success  
success  
[root@server ~]# systemctl restart postfix  
[root@server ~]# ss -ltnp | grep ':587'  
LISTEN 0      100        0.0.0.0:*        0.0.0.0:*      users:(("master",pid=19152,fd=17))  
[root@server ~]#
```

Смысл результата

openssl s_client установил зашифрованное соединение и вывел сертификат сервера. Команда EHLO показывает поддержку STARTTLS и AUTH.

Проверка AUTH LOGIN

```
ndbaranov@client:~
No ALPN negotiated
Early data was not sent
Verify return code: 18 (self-signed certificate)
---
250 CHUNKING
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
  Protocol : TLSv1.3
  Cipher   : TLS_AES_256_GCM_SHA384
  Session-ID: A29E97079ECFA401B7C6F342B9BCA14EDC86A21870FB4D88DF30EA159926E2A6
  Session-ID-ctx:
  Resumption PSK: 8CD79B101E687F2A8D47E6F302E98475A29BF7E3D241F59C7C37C0C7F28A20B217FE61E95D37B550AAF4
12AD02D73708
  PSK identity: None
  PSK identity hint: None
  SRP username: None
  TLS session ticket lifetime hint: 7200 (seconds)
  TLS session ticket:
0000 - c0 55 86 53 4c c5 8a 6b-79 70 ab 00 9a 94 ff ff .U.SL..kyp.....
0010 - 7f ce 12 d8 15 39 10 0e-33 f4 eb 4b d6 6a cd db ....9..3..K.j..
0020 - 63 93 df 44 37 17 6c 0d-c3 7a 5c 2f 81 fe 30 8b c..D7.l..z\/.0.
0030 - a7 07 30 c7 59 c1 66 d7-d1 9b b8 a5 67 f3 9f 32 ..0.Y.f....g..2
0040 - e1 68 a3 b8 3c c5 7f 03-69 62 91 e4 5b c3 64 66 .h.<...ib..[.df
0050 - ba 4f 77 8e dd b7 68 e0-dd c6 0a 9e b6 42 4e 53 .Ow...h.....BNS
0060 - 74 da bd ed 9e 7d a7 67-76 af d5 f7 3c 57 6f 80 t....}.gv....<Wo.
0070 - 5d f7 ee 66 0c 21 cb b7-eb 9c 1c e9 66 22 c4 c4 ].f.l.....f"..
0080 - 32 23 7b 05 60 37 83 21-90 f3 db 3c 5c 8b a3 ef 2#{`7.l...<...
0090 - 5a 53 5a b9 71 36 e6 25-b3 ae c9 62 ec a9 47 a5 ZSZ.q6.%...b..G.
00a0 - 9c 85 66 fe 05 3d 17 c0-7f 78 dc 1c c5 91 e2 3d ..f...X.....=
00b0 - 64 76 42 23 c5 ab 30 cb-28 15 96 74 48 73 90 57 dvB#.0.(.tHs.W
00c0 - 4e 7b 8d 2f c0 86 c7 db-bb f6 2e 81 e9 dd 3c 55 N{./.....<U

Start Time: 1788692594
Timeout : 7200 (sec)
Verify return code: 18 (self-signed certificate)
Extended master secret: no
Max Early Data: 0
---
read R BLOCK
EHLO test
250-server.ndbaranov.net
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITIME
250-DSN
250-SMTPUTF8
250 CHUNKING
AUTH PLAIN bmRiYXJhbm92AG5kYmFyYV5vdGxMjMONTY=
235 2.7.0 Authentication successful
```

Смысл результата

В тестовом SMTP-сеансе сервер принял корректные учётные данные. Ответ 235 означает успешную аутентификацию клиента.

Автоматизация расширенной почты

```
vagrant@client:~$ exit
logout
PS D:\work\ndbaranov\vagrant> cd D:\work\ndbaranov\vagrant
PS D:\work\ndbaranov\vagrant> notepad .\provision\server\mail.sh
PS D:\work\ndbaranov\vagrant> notepad .\provision\client\mail.sh
PS D:\work\ndbaranov\vagrant> vagrant validate
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
Vagrantfile validated successfully.
PS D:\work\ndbaranov\vagrant> vagrant provision server --provision-with "server mail"
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Running provisioner: server mail (shell)...
server: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell20260906-19116-ykjm09.sh
server: Provisioning advanced mail server
server: Install needed packages
server: Last metadata expiration check: 2:16:34 ago on Sun 06 Sep 2026 08:51:38 AM UTC.
server: Package postfix-2:3.8.5-10.el10_2.x86_64 is already installed.
server: Package dovecot-1:2.3.21-19.el10_2.1.x86_64 is already installed.
server: Package telnet-1:0.17-94.el10.x86_64 is already installed.
server: Package s-nail-14.9.24-12.el10.x86_64 is already installed.
server: Dependencies resolved.
server: Nothing to do.
server: Complete!
server: Configure firewall
server: Warning: ALREADY_ENABLED: smtp
server: success
server: success
server: Warning: ALREADY_ENABLED: pop3
server: Warning: ALREADY_ENABLED: pop3s
server: success
server: Warning: ALREADY_ENABLED: imap
server: success
server: Warning: ALREADY_ENABLED: imaps
server: success
server: Warning: ALREADY_ENABLED: smtp-submission
server: success
server: success
server: Configure Dovecot
server: Configure Dovecot LMTP and SASL sockets
server: Configure Postfix
server: Configure SMTP TLS
server: Configure submission port 587
server: Check configuration
server: Start services
server: Advanced mail server provisioning completed
PS D:\work\ndbaranov\vagrant>
```

Смысл результата

Сценарий advanced-mail-server последовательно применяет LMTP, SASL, TLS и submission. Его завершение подтверждает воспроизводимость настройки.

Результат

Postfix и Dovecot объединены через LMTP и SASL, а submission защищён TLS. Проверка openssl подтверждает шифрованное SMTP-соединение и объявление AUTH.